

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

Practical No : 9 and 10

Aim 9 : Set up Apache Spark locally and count the frequency of words in a text file.

Aim 10 : Filter rows in a CSV file using Spark DataFrames where a numeric column exceeds a threshold.

CODE 9 :

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("WordCount") \
    .master("local[*]") \
    .getOrCreate()

text_file = spark.sparkContext.textFile("input.txt")

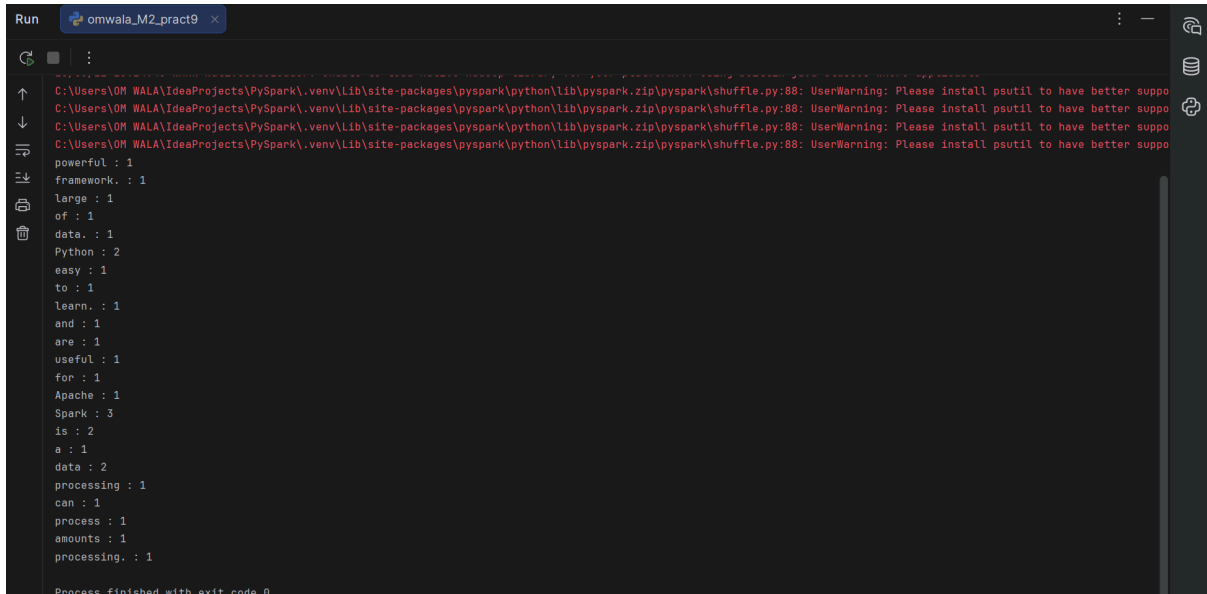
word_counts = (
    text_file
    .flatMap(lambda line: line.split())
    .map(lambda word: (word, 1))
    .reduceByKey(lambda a, b: a + b)
)

for word, count in word_counts.collect():
    print(word, ":", count)

spark.stop()
```

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OUTPUT :



```
Run omwala_M2_pract9 x
C:\Users\OM WALA\IdeaProjects\PySpark\.venv\Lib\site-packages\pyspark\python\lib\pyspark.zip\pyspark\shuffle.py:88: UserWarning: Please install psutil to have better support for monitoring the memory usage of the JVM.
C:\Users\OM WALA\IdeaProjects\PySpark\.venv\Lib\site-packages\pyspark\python\lib\pyspark.zip\pyspark\shuffle.py:88: UserWarning: Please install psutil to have better support for monitoring the memory usage of the JVM.
C:\Users\OM WALA\IdeaProjects\PySpark\.venv\Lib\site-packages\pyspark\python\lib\pyspark.zip\pyspark\shuffle.py:88: UserWarning: Please install psutil to have better support for monitoring the memory usage of the JVM.
C:\Users\OM WALA\IdeaProjects\PySpark\.venv\Lib\site-packages\pyspark\python\lib\pyspark.zip\pyspark\shuffle.py:88: UserWarning: Please install psutil to have better support for monitoring the memory usage of the JVM.
powerful : 1
framework : 1
large : 1
of : 1
data : 1
Python : 2
easy : 1
to : 1
learn : 1
and : 1
are : 1
useful : 1
for : 1
Apache : 1
Spark : 3
is : 2
a : 1
data : 2
processing : 1
can : 1
process : 1
amounts : 1
processing : 1
Process finished with exit code 0
```

CODE 10 :

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("FilterRows") \
    .master("local[*]") \
    .getOrCreate()

df = spark.read.csv(
    "students.csv",
    header=True,
    inferSchema=True
)

print("Original Data:")
df.show()

threshold = 75

filtered_df = df.filter(df["Marks"] > threshold)

print("Students with Marks greater than", threshold)
filtered_df.show()

spark.stop()
```

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OUTPUT :

```
Run omwala_M2_pract10 x
Setting default log level to WARN.
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/08/22 23:35:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using b
Original Data:
+-----+-----+-----+
|   Name|Subject|Marks|
+-----+-----+-----+
|    Om| Python|  98|
|Shreyash| Python|  85|
|   Krish| Python|  65|
|  Deepak| Python|  72|
| Krishna| Python|  55|
|   Sahil| Python|  78|
+-----+-----+-----+

Students with Marks greater than 75
+-----+-----+-----+
|   Name|Subject|Marks|
+-----+-----+-----+
|    Om| Python|  98|
|Shreyash| Python|  85|
|   Sahil| Python|  78|
+-----+-----+-----+

|
Process finished with exit code 0
```

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